Abstract—

I. Introduction

Electric vehicle fleets, such as e-taxis, have been deployed in metropolitan areas because of their lower maintenance costs and environmental benefits. However, the high demand that electric vehicles place on charging infrastructure can lead to long idle waiting time and increased peak electricity load, particularly for high-utilization e-taxi fleets. Battery-swapping stations (BSS) have therefore emerged as a promising alternative that enables rapid "refueling" by replacing depleted batteries with fully charged ones, while also offering the potential to reduce charging congestion by allowing battery charging to be scheduled flexibly over time [1], [2].

New challenges arise in the co-deployment of electric vehicle fleets and battery swapping stations because of their tight interaction. Electric vehicle fleets decide when and where to swap batteries to support daily operations and maintain passenger service quality. These decisions depend on the current and expected availability of fully charged batteries at each station, since limited availability can increase idle waiting time and reduce the taxi supply. From the station perspective, charging schedules should ensure enough fully charged batteries are ready when vehicles arrive in order to maintain revenue. Meanwhile, battery swapping stations act as large electricity consumers and may participate in power system demand response programs to lower system costs in exchange for financial incentives. As a result, station charging schedules must jointly satisfy vehicle swapping demand and power system demand response requirements, which creates a tightly coupled decision problem between fleet operations, station management, and grid interaction.

Previous research on battery swapping stations covers multiple aspects of planning and operation. On the infrastructure side, early studies [1], [3] develop centralized scheduling models that assign EVs to swapping stations while jointly minimizing vehicle detours and power generation cost under network power flow constraints. More recent work [4] focuses on e-taxi fleets, proposing demand-driven assignment and cost-aware swapping strategies to reduce passenger waiting time and operational cost. [5] studies the joint deployment of plug-in chargers and swapping stations for ride-hailing fleets. It shows that a hybrid charging-swapping network can improve fleet utilization under budget limits. On the operational side, [6] proposes modular battery-block rental and inventory management models that use vehicle trajectory data to optimize battery inventory and charging schedules under time-of-use electricity pricing.

Despite these advances, most existing studies treat EV fleet operations, i.e., taxi dispatching and battery swapping decisions, and swapping-station inventory management, i.e., battery charging schedules, as separate problems. There remains a lack of an integrated framework that coordinates the actions of EV fleets and battery swapping stations to jointly optimize taxi service quality and station revenue. Such a framework is required to capture the coupled interactions among mobility demand, battery availability, and power system infrastructure.

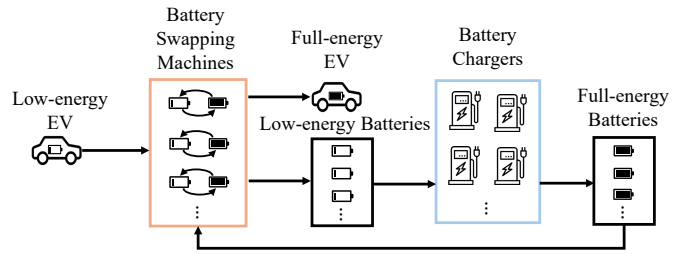

Fig. 1: Demonstration of battery swapping stations

II. System Design

A. Battery Swapping Stations

Figure 1 shows the architecture of a battery swapping station. It includes several components: (i) battery swapping machines that replace an EV's depleted battery pack with a fully charged one, (ii) depleted battery packs removed from EVs for recharging, (iii) battery chargers that restore these depleted packs to full charge, and (iv) fully charged battery packs ready to be installed in EVs requiring replacement. When an EV with a depleted battery arrives at the swapping station, and both an idle swapping machine and a fully charged battery pack are available, the machine replaces the EV's depleted battery pack. The removed battery packs are then scheduled for recharging to ensure sufficient availability for future swapping demands. In summary, a battery swapping station performs two primary functions: (i) replacing a depleted battery pack with a fully charged one, and (ii) recharging the removed battery packs to restore them to full capacity.

B. Design Framework

III. E-taxi Fleet Formulation

A. E-taxi Systems

Let us model the e-taxi fleet from the spatial-temporal dimensions. A city is divided into m areas, each equipped with at least an e-taxi battery swapping station. In addition, we break down a day into multiple time periods with the same length, each denoted by t. Let t' denote the current time period. We study the charging and rebalancing problem of the e-taxi fleet over the next T periods, i.e., spanning from

t' to $t'+T-1$, aiming to determine the battery demand at each station over time to support optimal service quality.

Throughout daily operation, an e-taxi's battery remaining energy ranges from 0% to 100%. We discretize the battery's state of charge into L levels. The energy level of an e-taxi at the start of time period t is denoted by E^t. The battery energy of an e-taxi evolves across time slots, and this change is captured by the following model. If an e-taxi is operating on the road during time period t, its remaining energy level at the start of period $t+1$ is given by $E^{t+1} = E^t - E^t_{i,i'}$, where $E^t_{i,i'}$ is the energy consumption when an e-taxi moves from area i to i' during period t. In contrast, if an e-taxi remains at a battery swapping station, its energy change is determined based on the availability of charged batteries and the competing demand from other e-taxis. A separate model will be introduced to characterize the energy dynamics at battery swapping stations.

E-taxi fleet managers are able to collect the real-time status of e-taxis supported by the installed sensing and communication devices. An e-taxi fleet's real-time status information contains the distribution of occupied and vacant e-taxis across the city, along with their remaining energy levels. Let $V^t_{i,l}$ ($O^t_{i,l}$) be the number of vacant (occupied) e-taxis, with the remaining energy l in area i at the start of slot t. It is noted that during period $t-1$, e-an taxi engaged in either searching passengers or swapping battery is regarded as vacant for the beginning of slot t.

Battery swapping & vehicle rebalancing actions: E-taxi fleet operators periodically assign vehicles to either passenger service or battery swapping based on the current e-taxi fleet status. This assignment process involves repositioning e-taxis for charging or passenger pickup as necessary. We define $X^{t,l}_{i,i'}$ as the number of e-taxis with remaining energy level l dispatched from area i to i' to provide passenger service at the beginning of period t. We define $Y^{t,l}_{i,i'}$ as the number of electric vehicles with remaining energy level l traveling from area i to i' for battery swapping at the beginning of slot t.

Fleet state transition: Based on each e-taxi's action during period t, we classify e-taxis into three groups: those swapping battery, those available for serving passengers, and those delivering passengers. Let $B^t_{i,l}$, $S^t_{i,l}$, and $U^t_{i,l}$ represent the amount of e-taxis with energy level l in area i that belong to each of three groups after being sent to swap battery or serve passengers. The relationship between the distribution of e-taxis before and after dispatching is

$$B^{t,l}_i = \sum_{i'=1}^{m} Y^{t,l}_{i',i},\ S^{t,l}_i = \sum_{i'=1}^{m} X^{t,l}_{i',i}. \tag{1}$$

The e-taxis assigned to serve passengers or already occupied with passengers determine the distribution of occupied e-taxis at the beginning of period $t+1$. Meanwhile, the distribution of vacant e-taxis at the start of period $t+1$ relies on the e-taxis' mobility pattern that are in any of the three categories during slot t. In summary, we describe $U^{t+1}_{i,l}$ and $O^{t+1}_{i,l}$ derived from $C^{t+1}_{i,l}$, $S^{t+1}_{i,l}$, and $O^{t+1}_{i,l}$ using the following model:

$$O^{t+1}_{i,l} = \sum_{i'=1}^{m} \tilde{P}^t_{i',i} S^t_{i',l+E^t_{i',i}} + \sum_{i'=1}^{m} P^t_{i',i} U^t_{i',l+E^t_{i',i}} \tag{2}$$

$$V^{t+1}_{i,l} = \sum_{i'=1}^{m} \tilde{Q}^k_{i',i} S^t_{i',l+E^t_{i',i}} + \sum_{i'=1}^{m} Q^t_{i',i} U^t_{i',l+E^t_{i',i}} + H^{t,l}_i, \tag{3}$$

where $P^t_{i',i}$ ($Q^t_{i',i}$) represents the likelihood of an occupied e-taxi repositioning from area i' to i during period t and being occupied (vacant) at the end of period t. Similarly, $\tilde{P}^t_{i',i}$ ($\tilde{Q}^t_{i',i}$) indicates the probability of a vacant e-taxi traveling from area i' to i during period t and still being occupied (vacant) at the end of period t. $H^t_{i,l}$ represents the amount of e-taxis with varying remaining energy for given the e-taxis assigned for battery swapping in area i during period t, which will be introduced later.

B. Battery Swapping Station Model

Battery swapping: Suppose battery swapping stations are widely deployed across the city to support the daily operation of e-taxi fleets. Let q_i denote the number of battery chargers at the swapping station in area i, used for recharging discharged battery packs. We define $M^{t,l}_i$ as the number of battery packs with remaining energy l at the beginning of period t at the swapping stations in area i. Note that only fully charged battery packs, i.e., $M^{t,L}_i$ can be used to replace the depleted battery packs in e-taxis. Let $\mu^{t,l}_i$ denote the number of e-taxis in area i during period t that arrive with energy level l and replace their depleted battery packs with fully charged ones through battery swapping. It is evident that $\mu^{t,l}_i$ is subject to three constraints: the number of e-taxis arriving at the swapping station with energy level l, the availability of fully charged battery packs, and the swapping station's service capacity p_i during period t. These constraints are expressed as follows:

$$\mu^{t,l}_i \le B^{t,l}_i,\ \ \sum_l \mu^{t,l}_i \le M^{t,L}_i,\ \ \sum_l \mu^{t,l}_i \le p_i. \tag{4}$$

Consequently, the number of e-taxis at the swapping station in area i with remaining energy l after battery swapping is given by $\forall\ l < L, H^{t,l}_i = B^{t,l}_i - \mu^{t,l}_i$, and $H^{t,L}_i = \sum_l \mu^{t,l}_i$. The first equation represents the number of e-taxis with energy level l that are not swapped due to the constraints defined in Eq. (4). The second equation gives the total number of e-taxis that successfully receive a fully charged battery and are thus assigned the highest energy level L after swapping.

Let $\hat{M}^{t,l}_i$ denote the number of battery packs with energy level l at the swapping station in area i that require full recharging after battery swapping. After battery swapping, $\hat{M}^{t,l}_i$ changes with the equation: $\hat{M}^{t,l}_i = M^{t,l}_i + \mu^{t,l}_i$. The first term accounts for the initial stock, and the second term adds the depleted battery packs returned after swapping.

Battery charging: The second task of a battery swapping station is to recharge the depleted battery packs, i.e., $\hat{M}^{t,l}_i$,

using q_i battery chargers. We assume that a battery pack can increase its energy level by $\hat{l}$ units if it is charged for one time period. If the number of depleted battery packs is less than or equal to the number of available charging poles, then all depleted packs can be charged. Otherwise, only q_i battery packs can be charged within a time period. Since the e-taxi fleet operators do not control the charging schedule at the swapping stations, we introduce $y_i^{t,l}$ to denote the number of battery packs with energy level l that are selected for charging in area i during period t. This quantity is subject to the following constraints:

$$y_i^{t,l} \leq \hat{M}_i^{t,l}, \quad \sum_l y_i^{t,l} \leq q_i. \tag{5}$$

Accordingly, at the beginning of period $t+1$, the number of battery packs that should be charged with energy level l at the swapping station in area i is updated as:

$$M_i^{t+1,l} = \hat{M}_i^{t,l} - y_i^{t,l} + y_i^{t,l-\hat{l}}. \tag{6}$$

C. E-taxi Fleet Optimization

Objectives: The key task of an electric taxi fleet is to optimize passenger service quality, which primarily involves maximizing the number of passengers served. This is achieved by strategically repositioning e-taxis either for battery swapping or to fulfill passenger requests. Fleet managers today collect vehicle trajectory data using mobile applications or embedded sensing and communication devices. These data are used to infer future passenger demand and improve taxi service quality. Let D_i^t denote the estimated number of passengers requesting taxi service in area i during time period t. The taxi supply, defined as the number of available electric taxis for serving passengers, is determined by the repositioning decisions, represented by $\sum_l S_{i,l}^t$. The taxi service quality is measured by the number of passengers served, given by

$$J_{service} = \sum_{t=t'}^{t'+T-1} \sum_{i=1}^{m} \min\{\sum_l S_i^{t,l}, D_i^t\}. \tag{7}$$

The repositioning of e-taxis for battery swapping or passenger service introduces additional overhead in the form of idle driving distance. This refers to the distance traveled by e-taxis without carrying passengers, which leads to energy waste and missed opportunities for passenger pickup. Therefore, a secondary objective of e-taxi repositioning is to minimize the total idle driving distance, formulated as:

$$J_{idle} = \sum_{i,i'} \omega_{i,i'} \sum_{t,l} (X_{i,i'}^{t,l} + Y_{i,i'}^{t,l}). \tag{8}$$

Constraints: The driving distance of an electric taxi during a given time period is limited by its maximum speed and the duration of travel. As a result, an e-taxi cannot be repositioned to a distant area for battery swapping or passenger service within a single time period. We define $\nu_{i,i'}^t \in \{0,1\}$ to indicate whether area i' is reachable from area i during period t. Specifically, $\nu_{i,i'}^t = 0$ if the travel is feasible within the time period, and $\nu_{i,i'}^t = 1$ otherwise. To ensure feasibility, the repositioning variables are constrained as follows:

$$X_{i,i'}^{t,l} \nu_{i,i'}^t = 0, Y_{i,i'}^{t,l} \nu_{i,i'}^t = 0. \tag{9}$$

Optimization problem: The task of fleet operators is to determine the decisions of battery swapping and vehicle rebalancing to maximize the service quality and minimize the idle driving distance due to repositioning. The problem is formulated as:

$$\max_{X,Y} \quad J = J_{service} + \beta J_{idle} \tag{10}$$

$$\textbf{s.t.} \quad \text{Eq. } (1) \sim (6), (9). \tag{11}$$

where β is a negative parameter to balance the weight between optimizing service quality and reducing idle driving distance.

D. Charging Task Generation

IV. Battery Charging Scheduler Design

We consider a city with m battery swapping stations, and there are p_j chargers deployed in the station j. In addition, we break down a day into multiple time periods with the same length, each denoted by t. Let t' denote the current time period.

Let $\tau_i = (A_i, C_i, D_i)$ represent the charging task for a battery pack, where A_i is the time slot when the battery pack is removed from an EV and becomes available for charging, C_i is the number of time slots required to fully charge the battery pack, and D_i the time slot by which the battery pack must be fully charged and ready for installation in an EV.

Because each battery pack is removed and charged at a designated swapping station, let Ψ_j denote the set of charging tasks for battery packs removed at station j. The entire charging task set to be executed on m battery swapping stations is denoted as $\Psi \triangleq \bigcup_{j=1}^m \Psi_j$. Since battery charging tasks arise during the daily operation of EV fleets, we assume that battery swapping stations plan for these tasks over the next T time slots given the current time slot t'. Specifically, for every $\tau_i \in \Psi$, the time indices satisfy $t' \leq A_i, C_i, D_i \leq t' + T - 1$.

Let $\mathbf{x}_i \in \{0,1\}^{1\times T}$ denote the charging schedule for task τ_i. If $x_i^t = 1$, the task is scheduled to charge during slot $t' + t$; otherwise, the corresponding battery pack remains uncharged. The execution of charging tasks is allowed to be preemptive and migratable. This means that when a battery pack is being charged, it can disconnect from its current charger to allow another battery pack, such as one with an earlier deadline, to begin charging. Afterward, the original battery pack can resume charging on a different charger within the same battery swapping station. To ensure that each charging task τ_i (i.e., charging the battery pack) is completed for C_i time slots before its deadline D_i, it is constrained that

$$\sum_{t=A_i}^{D_i-1} x_i^t = C_i. \tag{12}$$

Because fast charging requires high power, simultaneously charging multiple battery packs can cause a surge in the local power distribution network, potentially leading to grid instability. In addition, power demand peaks increase the cost of electricity supply. Therefore, it is important to minimize the peak charging power demand over a given time horizon, which is formulated as:

$$\min \max_{t \in [t', t'+T-1]} \sum_{\tau_i \in \Psi} x_i^t P + D_j, \tag{13}$$

$$\text{s.t.} \quad \sum_{t=A_i}^{D_i-1} x_i^t = C_i. \tag{14}$$

where P is the power of charging a battery pack.

V. Evaluation

VI. Related Work

A. Battery Swapping for Electric Taxi Fleets

Recent studies on battery swapping service for electric taxi fleets have focused on the interaction between vehicle dispatch decisions and swapping station utilization. Early work formulates swapping decisions as optimization problems driven by vehicle energy constraints and station service capacity, aiming to reduce operational cost and passenger waiting time []. For example, [4] proposes a mixed-integer optimization framework that jointly considers swapping cost, queuing delay, and environmental benefits, demonstrating the benefits of coordinated fleet-level swapping decisions. More recent efforts explicitly incorporate temporal dynamics and system-level coupling. Joint scheduling frameworks assign electric taxis to swapping stations based on state of charge, traffic conditions, and station congestion, revealing that decoupling dispatch and swapping decisions may lead to suboptimal fleet performance [7]. In addition, real-time guidance strategies have been investigated to adapt swapping decisions under time-varying demand and heterogeneous driver responses [8]. Network-level extensions further introduce multi-objective formulations that balance operational efficiency and environmental sustainability [9]. Overall, existing work highlights the importance of integrated dispatch and swapping decisions, while often treating station-level operations in a simplified or exogenous manner.

B. Operational Models for Battery Swapping Stations

A complementary line of research investigates battery swapping stations from an operational perspective, for example service capacity, inventory dynamics, and performance-cost trade-offs. [10] applies the dynamic operation models to characterize the evolution of battery inventory and charging processes under stochastic vehicle arrivals. It provides analytical insights into station utilization and waiting time performance. [11] further studies how battery availability and coordination among stations affect service reliability and operational efficiency with inventory-based models. In [12], the battery swapping and charging stations are formulated as stochastic service systems via queueing-theoretic models, which enables analysis of system stability and cycle time under different service configurations. In addition, [13] explicitly evaluates trade-offs between quality of service and station investment or operating cost. While these studies provide valuable insights into station-level dynamics, vehicle arrival processes are typically treated as exogenous and independent of fleet dispatch decisions.

C. Battery Charging Stations

The operation and control of electric vehicle charging stations have been extensively studied, with a primary focus on charging load management and grid-aware scheduling. Charging stations are commonly modeled as flexible electrical loads, and decentralized control strategies have been developed to regulate aggregate charging demand while preserving grid stability [14]. Optimization-based formulations further treat charging as a collection of deadline-constrained tasks to achieve load balancing and valley-filling objectives [15].

Survey studies highlight fundamental differences between charging and swapping infrastructures, noting that charging station models emphasize power delivery and load management over extended time horizons, whereas battery swapping stations introduce additional constraints related to battery inventory and service capacity [2]. As a result, charging station models generally do not consider vehicle dispatch decisions or battery inventory dynamics and are not directly applicable to systems that tightly couple fleet operations with battery swapping.

VII. Conclusions

References

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